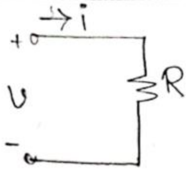
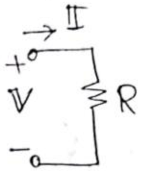
9.4 Phasor Relationships for Circuit Elements

Impedance Unit → Ω

Resistor:



$v = iR$

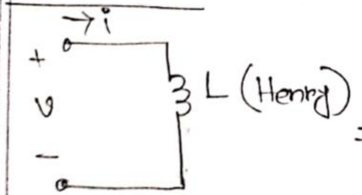


$\mathbb{V} = \mathbb{I}R$

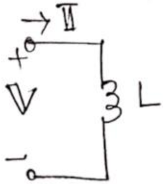
$Z_R = \frac{\mathbb{V}}{\mathbb{I}} = R$

$Z_R = R\angle 0^\circ$ Impedance of Resistance

Inductor:



$v = L\frac{di}{dt}$



$\mathbb{V} = Lj\omega\mathbb{I}$

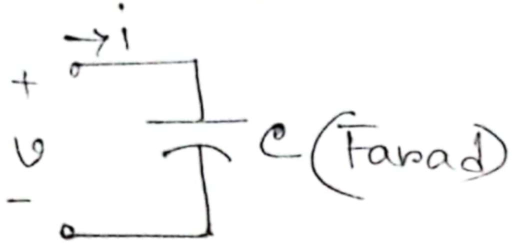
$\mathbb{V} = j\omega L\mathbb{I}$

$Z_L = \frac{\mathbb{V}}{\mathbb{I}} = j\omega L$

$Z_L = \omega L\angle 90^\circ$

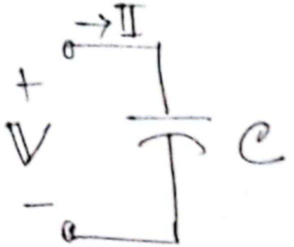
Impedance of Inductance

Capacitor:



$$i = C\frac{dv}{dt}$$

$\Rightarrow$



$$I = C\, j\omega V$$

$$V = \frac{I}{j\omega C}$$

$$Z_C = \frac{V}{I} = \frac{1}{j\omega C}$$

$$Z_C = \frac{-j}{\omega C}$$

$$Z_C = \frac{1}{\omega C}\angle -90^\circ$$

Impedance of Capacitance.

Example 9.8

$v = 12\cos(60t + 45°)$ V

$L = 0.1$ H

$i = ?$

$\mathbb{V} = 12\angle 45°$ V

$\mathbb{Z}_L = j\omega L = j \times 60 \times 0.1 = j6\ \Omega$

$\mathbb{Z}_L = \dfrac{\mathbb{V}}{\mathbb{I}}$

$\mathbb{I} = \dfrac{\mathbb{V}}{\mathbb{Z}_L} = \dfrac{12\angle 45°}{j6} = 2\angle -45°$ A

$i = 2\cos(60t - 45°)$ A

Practice Problem 9.8

$v = 10\cos(100t + 30°)$ V

$C = 50\mu F = 50 \times 10^{-6}$ F

$i = ?$

$\mathbb{V} = 10\angle 30°$ V

$\mathbb{Z}_C = \dfrac{1}{j\omega C} = \dfrac{1}{j \times 100 \times 50 \times 10^{-6}} = \dfrac{1}{j5 \times 10^{-3}}\ \Omega$

$\mathbb{I} = \dfrac{\mathbb{V}}{\mathbb{Z}_C} = 10\angle 30° \times j5 \times 10^{-3} = 50\angle 120°$ mA

$i = 50\cos(100t + 120°)$ mA

$Z_R = R\angle 0^\circ \ \Omega$

$Z_L = j\omega L$

$Z_C = \dfrac{1}{j\omega C}$

$\sin(\omega t \pm 180^\circ) = -\sin\omega t$

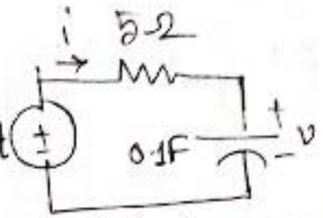
$\cos(\omega t \pm 180^\circ) = -\cos\omega t$

$\sin(\omega t \pm 90^\circ) = \pm \cos\omega t$

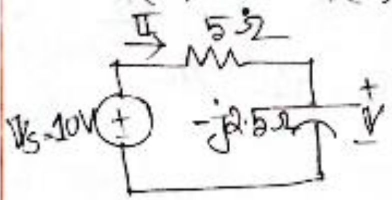
$\cos(\omega t \pm 90^\circ) = \mp \sin\omega t$

Ex 9.9

$v_s = 10\cos 4t$



$v(t) = ?$ $\quad i(t) = ?$



$Z_C = \dfrac{1}{j\omega C} = \dfrac{1}{j \times 4 \times 0.1} = -j2.5 \ \Omega$

$Z = Z_R + Z_C = (5 - j2.5)\ \Omega$

$I = \dfrac{V_s}{Z} = \dfrac{10\angle 0^\circ}{5 - j2.5} = (1.6 + j0.8)\ A = 1.789\angle 26.565^\circ \ A$

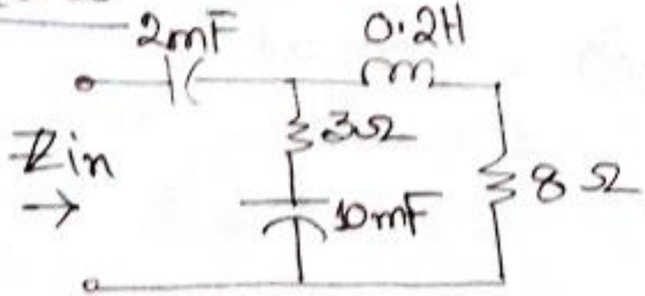
$i(t) = 1.789\cos(4t + 26.565^\circ)\ A$

$V = I Z_C = 1.789\angle 26.565^\circ \times (-j2.5) = 4.473\angle -63.435^\circ \ V$

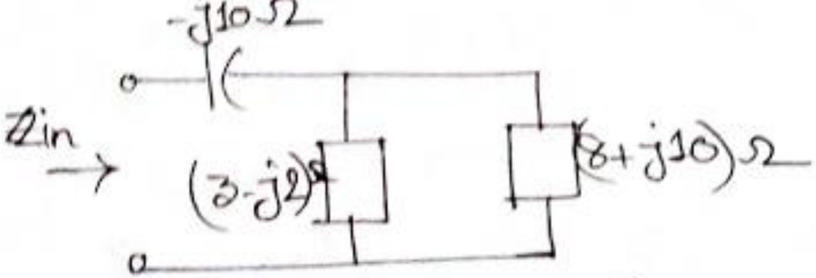
$v(t) = 4.473\cos(4t - 63.435^\circ)\ V$

P.P. 9.9

Ex 9.10



Given, $\omega = 50$ rad/s



$$Z_{in} = \frac{(8+j10)\times(3-j2)}{8+j10+3-j2} + (-j10)$$

$$= \frac{(8+j10)\times(3-j2)}{11+j8} - j10$$

$$= (3.22 - j11.07)\ \Omega$$

$\boxed{\text{Ex 9.11, P.P. 9.11}} \Rightarrow$ Do Practice

2mF:

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j\times 50\times 2\times 10^{-3}}$$

$$= -j10\ \Omega$$

$$3 + \frac{1}{j\omega C} = 3 + \frac{1}{j\times 50\times 10\times 10^{-3}}$$

$$= (3-j2)\ \Omega$$

$$8 + 0.2H = 8 + j\omega L$$

$$= 8 + j\times 50\times 0.2$$

$$= (8+j10)\ \Omega$$

Exercise

Exercise: 9.1, 9.2, 9.5, 9.6, 9.8, 9.9, 9.11, 9.13, 9.16, 9.20, 9.23, 9.24, 9.25, 9.35, 9.39, 9.42, 9.43, 9.44, 9.59, 9.60